

WEAK TOPOLOGIES HAVE DIFFERENTIABLE BLID PROPERTY

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SOURCE QUESTION

This packet concerns Genrich Belitskii and Victoria Rayskin, *Extension of differentiable local mappings on linear topological spaces*, arXiv:1812.11064 [1]. The paper defines the differentiable blid property for a linear topological space X : for every neighborhood U of 0, there is a differentiable map $H : X \rightarrow X$ which coincides locally with the identity and satisfies $H(X) \subset U$.

In conclusion, we would like to pose the following

QUESTION 5.8. Which linear topological spaces have differentiable blid property?

This question was not answered even for Banach spaces.

6. Applications

Local linearization and normal forms are convenient simplification of complex dynamics. In this section we discuss differentiable linearization on Banach spaces.

The source asks on PDF page 6:

Question 5.8. Which linear topological spaces have differentiable blid property?

It immediately adds that this question was not answered even for Banach spaces.

RESULT

We give a broad positive class outside the normed setting.

Theorem 1. *Let X be a real vector space endowed with the weak locally convex topology $\sigma(X, \mathcal{F})$ generated by a family $\mathcal{F}$ of linear functionals on X . Then X has C^∞ differentiable blid property. More precisely, for every neighborhood U of 0 there is a C^∞ map $H : X \rightarrow X$ such that $H = \text{id}$ on a neighborhood of 0 and $H(X) \subset U$.*

Corollary 1. *For every index set I , the product space $\mathbb{R}^I$ with the product topology has C^∞ differentiable blid property.*

PROOF INTUITION

In a weak topology, every neighborhood of zero is controlled by finitely many linear coordinates. One can therefore reduce the problem to a finite dimensional cutoff. If $\Phi : X \rightarrow E$ is the finite coordinate map defining a smaller neighborhood $V \subset U$, choose a smooth bounded local identity $b : E \rightarrow E$ whose image lies in the coordinate set $\Phi(V)$. A linear section $R : E \rightarrow X$ lets us replace only the finite coordinate component of x :

$$H(x) = x - R\Phi x + Rb(\Phi x).$$

Then H is identical when Φx is small, but all values of $\Phi(Hx)$ lie in the desired finite coordinate neighborhood.

PROOF OF THEOREM 1

Let U be a neighborhood of 0 in $\sigma(X, \mathcal{F})$. By the definition of the weak topology, there are functionals $f_1, \dots, f_n \in \mathcal{F}$ and positive numbers $\varepsilon_1, \dots, \varepsilon_n$ such that

$$V = \{x \in X : |f_j(x)| < \varepsilon_j, 1 \leq j \leq n\} \subset U.$$

If $n = 0$, take $H = \text{id}$. Otherwise set

$$\Phi : X \rightarrow \mathbb{R}^n, \quad \Phi x = (f_1(x), \dots, f_n(x)),$$

and let $E = \Phi(X)$, a finite-dimensional subspace of $\mathbb{R}^n$. Put

$$W = \{y \in E : |y_j| < \varepsilon_j, 1 \leq j \leq n\}.$$

Choose a Euclidean norm $|\cdot|$ on E . There is $\rho > 0$ such that $\{y \in E : |y| < \rho\} \subset W$.

Choose a scalar C^∞ function $\lambda : [0, \infty) \rightarrow [0, 1]$ such that $\lambda(s) = 1$ for s near 0 and

$$\lambda(s)\sqrt{s} < \rho \quad (s \geq 0).$$

For instance, take $\lambda = 1$ near 0, make λ compactly supported, and shrink its tail if necessary. Define

$$b : E \rightarrow E, \quad b(y) = \lambda(|y|^2)y.$$

Then b is C^∞ , it coincides with the identity near 0, and $b(E) \subset W$.

Since $\Phi : X \rightarrow E$ is onto, choose an algebraic linear section $R : E \rightarrow X$, so $\Phi R = \text{id}_E$. This section is continuous: every functional $g \in \mathcal{F}$ gives $g \circ R$, a linear functional on the finite-dimensional space E , hence a continuous map on E . Define

$$H : X \rightarrow X, \quad H(x) = x - R\Phi x + Rb(\Phi x).$$

Then

$$\Phi(Hx) = \Phi x - \Phi R\Phi x + \Phi Rb(\Phi x) = b(\Phi x) \in W,$$

so $H(X) \subset \Phi^{-1}(W) = V \subset U$.

Also $H(x) = x$ whenever Φx lies in the neighborhood of 0 on which $b = \text{id}_E$. Therefore H is locally identical to the identity.

It remains only to record differentiability. The map H is the identity plus the finite-dimensional cylinder map

$$R \circ (b - \text{id}_E) \circ \Phi.$$

For $k \geq 2$,

$$D^k H(x)(h_1, \dots, h_k) = R D^k b(\Phi x)(\Phi h_1, \dots, \Phi h_k),$$

and

$$DH(x)h = h + R(Db(\Phi x) - I_E)\Phi h.$$

These are continuous multilinear maps in the weak topology, because they are compositions of continuous linear maps with finite-dimensional smooth derivatives. Thus H is C^∞ in the standard differentiability senses used in the source paper, including bounded and compact/Hadamard differentiability. This proves the theorem.

PROOF OF THE COROLLARY

The product topology on $\mathbb{R}^I$ is the weak topology generated by the coordinate functionals $x \mapsto x_i$, $i \in I$. The corollary follows from Theorem 1.

VERIFICATION AND NOVELTY NOTES

The proof is direct and uses no computation. The local duplicate check found no prior packet for arXiv:1812.11064 or for the weak-topology blid subcase. The closest run memory is an unpromoted attempt for arXiv:1706.08513 on the harder question of smooth blid maps on non-even ℓ_p . Local source search found the earlier blid-map paper arXiv:1706.08513 and the later Whitney-extension paper arXiv:1910.14248; neither records this weak-topology subcase. Web searches for the exact source question and phrases such as “differentiable blid property weak topology” and “blid map weak topology” did not surface a later answer.

LIMITATIONS AND HUMAN REVIEW RECOMMENDATION

This is a partial result. It does not classify Banach spaces, and it does not address strong locally convex topologies defined by infinite-dimensional seminorms. Review should focus on the differentiability claim for weak topologies and on the continuity of the finite-dimensional section $R : E \rightarrow X$.

REFERENCES

- [1] G. Belitskii and V. Rayskin, *Extension of differentiable local mappings on linear topological spaces*, arXiv:1812.11064.
- [2] G. Belitskii and V. Rayskin, *A new method of extension of local maps of Banach spaces. Applications and examples*, arXiv:1706.08513.
- [3] G. Belitskii and V. Rayskin, *Whitney extension problem for Banach spaces*, arXiv:1910.14248.